APV-MaGa

Agrivoltaic Monitoring Platform

User Manual

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1. Introduction to APV-MaGa

APV-MaGa is a monitoring dashboard developed by **Saratech** for the collection and real-time surveillance of agrivoltaic installations, anywhere in the world.

Every 10 minutes the platform automatically collects the measurements of the sensors connected to **ThingsBoard** (soil moisture, valves, flow, tank) and of the **LI-COR / HOBOnet** stations (weather, PAR). It also accepts data from IoT devices (ESP32, Arduino) via a dynamic REST API. It also allows manual entry by field agents for important parameters not collected by sensors — such as plant height, leaf count, harvest yield in kg or monetary value.

APV-MaGa is designed to adapt to all contexts: managing a single site or centralizing multiple sites across different countries, with a multi-user and multi-role interface.

Key Features

Real-time
Display refreshed automatically, without reloading the page

Automatic sources
ThingsBoard, LI-COR (HOBOnet) and REST API for ESP32

Charts
Chart.js visualization with period filter

Manual entry
Structured forms for field data

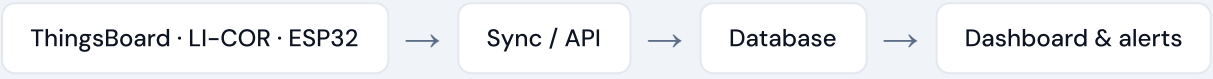
Export
CSV and Excel by category or full site

Multi-user
Admin, Agent, Observer roles

E-mail alerts
A message when part of a site stops sending data

Computed values
Tank volume, daily water use

System Architecture



2. Login & Roles

Login

Access the platform via your browser and enter your credentials.

```
https://<your-domain>/login
```

No account yet? The registration page records a request: the account stays pending until an administrator approves it, and you get an e-mail as soon as it is activated.

User Roles

Admin
Full access — manage sites, users, categories, parameters, charts

Agent
Assigned sites — view, manual entry, CSV/Excel export

Observer
Assigned sites — view and export only

The Agent or Observer role is also chosen site by site: the same person can enter data on one site and only view another.

3. Dashboard — Overview

The home page displays all your sites with their real-time key indicators.

Interface Elements

Element	Description
Sidebar	Sites list with colored status (green=active, orange=alert). On a computer it collapses to icons; the choice is remembered.
Live badge	Green = data is arriving; orange = the latest data is too old; grey = no data. Hover it to see the time of the last reading.
Topbar	Title, UTC clock, period filter 1H/6H/24H/7D/Custom (exact dates), CSV/Excel buttons
Site Card	Indicators available for the site (soil moisture, tank, weather, PAR…) and its status: Normal, Warning, Critical, No recent data or No data
Sparkline	Mini-chart of the site's main indicator
Cross-Site Trends	Comparative charts across all sites

The display refreshes automatically without reloading the page: every minute here, every 30 seconds on a site's page.

Custom Date Range

In addition to the 1H/6H/24H/7D buttons, the "Custom" button opens a date picker (start/end) to view any past period across charts, raw data, and exports. Once applied, the chosen period shows as a blue badge (e.g. "Jul 1 → Jul 5") next to the buttons — click the badge's × to clear it and return to the default period.

4. Site Detail

Click on a site in the sidebar to access its detailed dashboard.

Available Tabs

Tab	Content
`Solar, Water...`	Tabs per category configured in the Admin Panel
`Raw Data`	Raw data grouped by timestamp with filter and pagination
`Manual Input`	Manual entry forms (if manual parameters configured)
`API Key`	API key and endpoint documentation for ESP32

KPI Cards

Sensor
Latest value received from ESP32 sensor

Manual
Latest manually entered value

Data status (No data / Stale)

Each card also shows how fresh its data is. If a parameter has never received data, or if the latest data is too old, a badge appears directly on the card:

No data
No data has ever been received for this parameter

Stale (Xm)
Data exists but is older than the category's threshold (60 min for ThingsBoard sensors, 12 h for LI-COR stations, 5 min otherwise)

Warning
The value has crossed the parameter's configured "Warning threshold"

Critical / Out of range
The value has crossed the "Critical threshold", or falls outside the configured Min/Max range

Hover the badge to see the exact timestamp of the latest data ("Last data: X ago"). The "Stale" threshold is configurable per category in Admin Panel → Categories → Edit. Warning/Critical thresholds and the Min/Max range are configured per parameter in Admin Panel → Parameters.

Grouped parameters: on the dashboard, parameters sharing the same group (configured in Admin Panel → Parameters → Manage Groups) are displayed boxed together in a colored panel, instead of as separate cards.

Switches & actuators (valves, pumps, fans)

Some parameters (type `Switch`) display as an ON/OFF toggle instead of a numeric value. There are two variants:

Read-only

The toggle is grayed out and **not clickable**. Its ON/OFF state automatically reflects the latest data sent by the sensor. Badge: "**Auto (sensor)**".

Controllable

The toggle is **clickable** (Admin and Agent only — not Observer). Clicking opens a confirmation popup, then sends a command to the device.

TURNING A CONTROLLABLE SWITCH ON/OFF

1. Click the toggle on the parameter's card (e.g. "Valve 1")
2. A confirmation popup appears: "Turn ON/OFF [name]?"
3. Click **Confirm** to send the command, or **Cancel** to abort (the toggle reverts to its previous state)
4. The device applies the change on its next check-in (a few seconds to a few minutes depending on the device)

SYNC INDICATORS

Below each controllable switch, a badge shows whether the command was actually applied by the device:

Badge	Meaning
`Synced`	The device confirmed the same state that was requested.
`Pending…`	The command was sent but the device hasn't confirmed yet — wait for its next check-in.
`No device report`	No data has ever been received from the device for this parameter — cannot confirm.
`Stale`	The confirmed state matches the command, but the last report is too old — the device may be offline.

⚠ If the device appears offline, the confirmation popup shows a warning before sending the command — it will still be queued and applied once the device reconnects.

5. Manual Input

Allows agents to enter field data that does not come from automatic sensors.

How to Enter Data

1. Click the `Manual Input` tab in the tab bar
2. Select the **date** of the field measurement
3. Choose the **plot** : reference plot or APV plot, so plots can be compared in Analysis
4. Enter **numeric values** for each parameter
5. Add optional **notes** for context
6. Click `+ Save` to save
7. The page automatically reloads on **Manual Input**
8. **KPI cards** update immediately

> ⚠ The Manual Input tab only appears if the site has at least one parameter configured with input_type = manual.

6. Raw Data & Export

Raw Data Features

Feature	Description
Period filter	1H, 6H, 24H, 7D, 30D — without reloading the page
Grouping	1 row per sensor send timestamp
Type badge	Sensor (green) or Manual (blue)
Pagination	20 records per page
Per-category export	CSV or XLS for each category

Export Options

CSV Category
Category data as CSV (UTF-8)

XLS Category
Category data as Excel (colored headers)

All CSV Site
All categories in one CSV file

All Excel Site
All categories as separate Excel sheets

The topbar's CSV/Excel buttons export according to the active period (1H/6H/24H/7D or the custom range chosen via "Custom").

6.3 Saved Records

The **Saved Records** table appears in the Raw Data tab and in each category tab with grouped manual parameters. It displays entries by group with dynamic columns.

Element	Description
Group filter	Dropdown to display a specific group (e.g. Yield Data, Plant Measurements)
Counter	Number of records for the selected group
Dynamic columns	Columns correspond to the parameters of the selected group
CSV / XLS	Exports only the data for the currently selected group

7. ESP32 API

Dynamic REST API for connecting your ESP32/Arduino sensors. ThingsBoard sensors and LI-COR stations do not need it: they are synchronised automatically (see "Sources & Alerts").

Available Endpoints

POST	`/api/sensors/{site}/{category}`	Send data
GET	`/api/sensors/{site}/{category}/latest`	Latest reading
GET	`/api/sensors/{site}/status`	Global status
GET	`/api/commands/{site}/{category}`	Switch commands (valves, pumps…)
GET	`/api/sensors/{site}/{category}/history`	Raw data over a date range (research)
GET	`/api/sensors/{site}/{category}/stats`	Aggregated statistics (average, min, max…)

Authentication

```
X-API-Key: apv-xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx
```

The key is only accepted in this header, never in the URL. Always use an https:// address.

Request Example

```
POST /api/sensors/your-site-slug/solar
Content-Type: application/json
X-API-Key: apv-xxxxxxxxxxxxxxxxx

{
  "solar_output": 4.2,
  "solar_irradiance": 890,
  "panel_temperature": 54.1
}
```

Arduino/ESP32 Code

```
HTTPClient http;
http.begin("https://domain.com/api/sensors/site-slug/solar");
http.addHeader("X-API-Key", "apv-xxxxxxxxx");
http.addHeader("Content-Type", "application/json");

String body = "{\"solar_output\":" + String(power) +
              ",\"solar_irradiance\":" + String(irr) + "}";
int code = http.POST(body);
http.end();
```

Switch commands (valves, pumps…)

If the category contains `data_type=switch` and `control_type=controllable` parameters (e.g. `valve_1`), the ESP32 must poll this endpoint regularly to know which state to apply:

```
GET /api/commands/your-site-slug/irrigation
X-API-Key: apv-xxxxxxxxxxxxxxxxx

{
  "commands": {
    "valve_1": { "desired_state": 1, "reported_state": 0 }
  }
}
```

The ESP32 sets the relay according to `desired_state`, then sends its real state (`valve_1: 1`) in the next `POST /api/sensors/...` — this automatically updates `reported_state`.

> Parameter slugs must exactly match those configured in Admin Panel → Parameters.

Research & Analysis — historical data

For researchers and statistical analysis, two endpoints let you extract data over a date range without using the dashboard's CSV export.

RAW DATA OVER A DATE RANGE

```
GET /api/sensors/{site}/{category}/history?from=2026-05-01&to=2026-05-08
X-API-Key: apv-xxxxxxxxxxxxxxxxx
```

Returns each reading (1 row = 1 parameter + 1 timestamp), ideal for importing into Python/pandas or R. Maximum 90 days per request; beyond that, use the aggregated statistics below.

AGGREGATED STATISTICS (LONG-TERM TRENDS)

```
GET /api/sensors/{site}/{category}/stats?period=day&from=2026-05-01&to=2026-05-31
X-API-Key: apv-xxxxxxxxxxxxxxxxx
```

Returns, for each numeric parameter, the average/min/max/sum grouped by `period` (`hour`, `day`, `week`, or `month`). Useful for visualizing a trend over several months without downloading every reading.

Add `¶ms=solar_output,solar_irradiance` to limit to specific parameters (otherwise all parameters are included).

8. Administration

⚠ This section is for administrators only.

User Management

Admin Panel → Users

Action	Description
`+ Add User`	Create a user with role, status and country
`Approve`	Approve a user pending activation; they are told by e-mail. Each new registration request is e-mailed to the administrators.
`Suspend`	Temporarily suspend access
`Activate`	Reactivate a suspended user
`Delete`	Permanently delete. The last active administrator cannot be deleted, suspended or demoted.

8.2 Site Users

Admin Panel → Sites → `Users` — Manage users for each site. A user can be assigned to multiple sites with different roles.

Action	Description
`Add User`	Assign a user to the site with a role (Agent or Observer)
`Remove`	Remove a user — they immediately lose access to the site
`Agent`	Views, enters manual data, accesses API key
`Observer`	View and export only — no manual input or API Key

8.3 Site Management

Admin Panel → Sites → + New Site

Check "Auto-create all categories" to automatically generate Solar, Water, Irrigation, Weather and Agriculture with their default parameters.

Action	Description
Edit	Updates the site's name, country, assigned agent, status, coordinates, capacity, area and description, as well as its data sources (ThingsBoard prefix, LI-COR station, sensor locations) and its tank dimensions. The API key stays unchanged so already-configured IoT devices keep working.

Action	Description
Duplicate	Creates a full copy of the site (categories, parameters, parameter groups and charts) with a new name, slug and API key. Useful for rolling out a new site with the same structure as an existing one. No historical data (readings, actuator commands) is copied.

Data sources

Admin Panel → `Data sources` lists what the ThingsBoard and LI-COR accounts expose, and the site each source feeds. One click links a source to an existing site or creates the site; its categories, parameters and data appear at the next sync (10 minutes at most). A source can feed one site only.
In Admin Panel → Sites, a badge per site shows the state of each sync: **ok**, **late** (no successful sync recently) or **failed** (hover it to read the error).

8.4 Categories & Parameters

Parameters define the data collected by each category. Each parameter corresponds to a specific measurement (e.g. temperature, water level, agricultural yield).
Path: `Admin Panel → Sites → Categories → Parameters → + Add Parameter`

> **"No data" / "Stale" threshold:** click `Edit` on a category to set the **"Offline threshold (minutes)"** field (5 by default; 60 for categories fed by ThingsBoard and 720 for those fed by LI-COR, set automatically when the source is linked). If a sensor in this category sends nothing for longer than this delay, its parameters show "Stale" on the dashboard.

Field	Possible Values	Description
Name	`Free text`	Name displayed on KPI cards and charts (e.g. Ambient Temperature)
Slug	`Unique snake_case`	Identifier used in ESP32 API as JSON key (e.g. temperature_c). Must be unique within the category.
Unit	`Free text`	Measurement unit displayed (e.g. °C, kW, %, m, kg). Appears on KPI cards and exports.
Input type	`sensor / manual`	sensor = data sent automatically by ESP32 via API. manual = data entered manually by a field agent. Forced to sensor if Data type = Switch.
Data type	`float / integer / string / boolean / switch`	float = decimal number (e.g. 4.2 kW). integer = whole number (e.g. 12 plants). string = text (e.g. Good condition). boolean = Yes/No. switch = ON/OFF toggle (valve, pump, fan…).
Control type	`readonly / controllable`	Only visible if Data type = Switch. readonly = display automatically follows the sensor. controllable = clickable toggle from the dashboard.
Show on dashboard	`Yes / No`	If enabled, the parameter appears as a KPI card at the top of the tab with its latest value.
Show on chart	`Yes / No`	If enabled, the parameter is included in the category trend charts.

Field	Possible Values	Description
Min / Max	`Number (optional)`	Physically valid value range. A value outside it triggers the "Out of range" badge (red).
Warning / Critical threshold	`Number (optional)`	Two independent alert levels — orange (Warning) and red (Critical) badge on the parameter's card.
Alert direction	`below / above`	below (default) = alert when the value drops to/under the threshold (e.g. tank level). above = alert when the value climbs to/over the threshold (e.g. temperature).
Group	`Selected from a list`	Attaches the parameter to a group managed in Admin Panel → Parameters → Manage Groups (see below). Parameters in the same group are boxed together on the dashboard, and grouped into one form on Manual Input.
Zone	`Selected from a list`	Where the parameter is measured: Reference, APV field, Under PV, Between PV or Module edge. The Analysis page compares the same quantity between zones. Set automatically for ThingsBoard and LI-COR sensors.

Example: To manually measure plant height: Slug = `plant_height`, Type = `float`, Unit = `cm`, Input type = `manual`, Group = `Plant Measurements`.

Managing Parameter Groups

Path: `Admin Panel → Sites → Categories → Parameters → Manage Groups`. This panel (collapsed by default — click the button to open it) lets you create, rename, recolor, reorder and delete a category's groups.

Action	How
Create	Enter a name and pick a color (a distinct color is suggested automatically) in the form at the bottom of the panel.
Rename / recolor	Edit the name or color directly on the group's card — saves automatically.
Reorder	Drag and drop a group's card to change its display order on the dashboard.
Delete	The group is removed but its parameters are not deleted — they simply become "ungrouped" again.

A group containing a single parameter displays as a normal card, without a box — the colored box only appears once 2 or more parameters share the same group.

8.5 Creating Charts

Charts are configured by the administrator for each category. They allow visualizing parameter evolution over time.

Path: `Admin Panel → Sites → Categories → Charts → + Add Chart`

CHART CONFIGURATION

Field	Values	Description
Title	`Free text`	Title displayed at the top of the chart (e.g. Solar Output vs Irradiance)
Type	`line / bar / area`	line = smooth curve. bar = vertical bars, suited to daily totals (rain, water use). area = curve with filled zone below.
Width	`full / half / third`	full = full width. half = 2 charts side by side. third = 3 charts side by side.
Height	`Pixels (e.g. 220)`	Chart height in pixels. 220px recommended for half/third, 280px for full.
Show legend	`Yes / No`	Show or hide the data series legend below the chart.

ADDING PARAMETERS TO CHART

After creating the chart, click `+ Add Parameter` to add data series. Each added parameter becomes a curve or bar on the chart.

Option	Values	Description
Parameter	`Dropdown list`	Select the parameter to display (must be float or integer).
Color	`Hex code (e.g. #1d6ed8)`	Curve or bar color. Choose contrasting colors to differentiate series.
Axis	`left / right`	left = left Y axis (main). right = right Y axis (secondary). Use right when two parameters have very different scales (e.g. kW and %).
Dashed	`Yes / No`	If enabled, the curve displays as dashed. Useful for a reference value or theoretical threshold.
Fill	`Yes / No`	If enabled, the area below the curve is colored (area effect). Recommended for only one series per chart.

Example: "Output & Irradiance" chart — add `solar_output` (blue, left axis, fill enabled) and `solar_irradiance` (orange, right axis, dashed), line type, full width.

⚠ If no chart is configured for a category, an automatic chart is generated with all numeric parameters of the category.

9. Data Sources & Alerts

Where the displayed data comes from, how fresh it is, and when the platform warns you.

Where the data comes from

Source	Measurements	Delay
ThingsBoard	Soil moisture, valves, flow, tank level and turbidity	Collected automatically every 10 minutes. The sensors report every 15 to 20 minutes.
LI-COR (HOBOnet)	Weather station (temperature, humidity, wind, rain, pressure) and PAR sensors	Stations upload their measurements in batches, every 6 to 10 hours: data appears with that delay, at its original timestamp. Values in 10-minute steps.
REST API (ESP32)	Any sensor-type parameter	Immediate, each time the device sends.
Manual entry	Field measurements (plants, harvest…)	When the form is saved.

Computed values

Tank volume	Computed from the measured water height and the site's own tank (radius and number of tanks, in Admin Panel → Sites → Edit). Without dimensions, no volume is shown rather than a wrong one.
Daily water use	Water used each day, computed from the flow meter and shown as bars.

A broken line means no data was received during that interval: the platform does not join the points on either side of a gap.

E-mail alerts

When a category of a site (Irrigation, Water, Weather…) sends nothing for longer than its **Offline threshold**, an e-mail goes out; a second one announces that data is back.

Who is told	Active administrators, the active users assigned to the site, and its owner, when their "Receive offline alerts" box is ticked (Admin Panel → Users). Ticked by default for administrators and agents, unticked for observers.
How often	One message when it stops, one when it resumes. Nothing is repeated during the outage. Categories of one site that change together share one message.
Single sensor	A sensor that goes quiet while the rest of its category still reports (a dead flow meter among working probes) is announced too, after three times the category threshold: 3 h for ThingsBoard, 36 h for LI-COR. Shorter gaps are routine on radio links and are not announced.
Not announced	A category that never sent data, a deactivated category or site, manually entered parameters, and silences caused by the platform itself not reaching ThingsBoard or LI-COR.

⚠ Value thresholds (Warning / Critical) show on the cards but are not sent by e-mail.

10. Analysis and comparison

The **Analysis** page (sidebar) compares measurements and explains the results in plain language. It answers the project's research questions, for example: how much light do the panels take from the crops? Does the soil stay wetter under the panels? Is the yield different?

Four ways to compare

Mode	What is compared	Example
`Zones of a site`	One quantity between the zones of a site: reference, under the panels (Under PV), between rows (Between PV), module edge, APV field. For field measurements: between the reference plot and the APV plot.	Light (PAR) at UTG: Under PV receives about 42 % of the reference light.
`Sites`	One quantity between several sites, in the same zone. The first site ticked is the baseline.	Reference air temperature at UTG, AfriFarm and Fass.
`Two periods`	One quantity at the same place over two periods, compared day by day from the start of each.	Soil moisture before and after planting; dry season against rainy season.
`Any parameters`	2 to 4 parameters from any of your sites on one chart, with one axis per unit. The first two are related by a scatter plot and a correlation coefficient.	Soil moisture and air temperature.

Running an analysis

1. Choose the mode, then the site (or sites, or parameters) and the quantity. Only combinations that are actually measured are offered.
2. Choose the period (7, 30 or 90 days, or exact dates; 92 days at most) and the step: hourly or daily values.
3. Click **Analyse**. The results appear below the form.
4. To start again, click **New analysis**: the screen is cleared, no data is deleted.

Reading the results

Findings	The results in plain language. Every figure comes from the summary table: nothing is estimated or made up.
Summary	For each series: number of sensors, mean, minimum, maximum, standard deviation, share of data available, and difference from the first series (the baseline).
Time series	The series on one chart. A broken line marks a period without data.
Difference from the baseline	Each series minus the baseline, hour by hour or day by day. Above zero: higher than the baseline.

Average day	The mean value at each hour of the day (UTC): shows when in the day the series differ.
Relationship	"Any parameters" mode: one point per hour or day, the trend line and the coefficient r, from -1 to 1. A correlation shows that two series move together, not that one causes the other.
Method and data quality	How values are computed, over which periods, and with what share of data available.

How values are computed

- Readings are averaged per hour or per day, in UTC. Rain and irrigation water are summed.
- A zone with several sensors (two soil probes, for example) is represented by their average, each sensor counting equally.
- Differences are computed only on the hours or days where both series have data: an outage on one side does not bias the result.
- With daily values, light (PAR) becomes the daily light integral (DLI, in mol/m²/day): mean PAR of the day × 0.0864.
- Differences in % are given only where they make sense (light, wind, rain, yield…), not for temperatures.
- A caution is shown when less than 70 % of the period is covered by data.

Downloading and sharing

Data (CSV)	The aligned values of every series, one row per hour or day, to continue the analysis in Excel, R or Python.
Report (PDF)	An A4 report with the findings, table, charts and method. The print dialog opens: choose "Save as PDF".
Link	The page address holds every choice of the analysis: copy it so a colleague sees exactly the same analysis (they need access to the same sites).

To compare yields and plant measurements between plots, give the plot ("Plot" field) with each entry in the Manual Input tab.

⚠ A comparison by zone needs each sensor to have its zone. It is set automatically for ThingsBoard and LI-COR sensors; for others, an administrator sets it in Admin Panel → Parameters → Edit ("Zone" field).

11. FAQ

My sensor sends data but it doesn't appear?

Check that category and parameter slugs match exactly those in the Admin Panel. The API key must be for the correct site. Check the site's API Key tab.

How to add a new category?

Admin Panel → Sites → click "Categories" → "+ Add Category". Then add parameters via "Parameters".

Charts show "No data"?

No data has been received for the selected period. Change the period (1H → 24H → 7D) or send data via the API.

The Manual Input tab doesn't appear?

The tab only appears if the site has at least one parameter with input_type = manual. Configure it in Admin Panel → Parameters.

How to connect multiple ESP32 to the same site?

Each ESP32 can send to a different category using the same API key. ESP32-1 → /solar, ESP32-2 → /water, etc.

How to change my password?

Click your name at the bottom of the sidebar → My Profile → Change Password.

Do manual data appear on charts?

Yes, if the parameter has data_type = float or integer. Manual data appears in trend charts and in Raw Data.

The Saved Records table does not appear?

The Saved Records table only appears if the category has manual parameters with a group name. Configure the Group name field in Admin Panel → Parameters.

How to assign multiple users to a site?

Admin Panel → Sites → click the Users button for the site → Add User. You can assign as many users as needed with different roles (Agent or Observer).

I clicked a switch but nothing happens on site?

The command is saved immediately, but the device only applies it on its next check-in ("Pending…" badge). If the badge stays "Pending…" for a long time or shows "No device report"/"Stale", the device is likely offline or misconfigured.

What's the difference between a "readonly" and "controllable" switch?

readonly = simple ON/OFF indicator, not clickable, reflects the sensor's state. controllable = clickable toggle that sends a command to the device, with a sync badge (Synced/Pending/Stale).

A card shows "No data" or "Stale" — what should I do?

"No data" = no data has ever been received for this parameter — check the slug and API key. "Stale" = data exists but is older than the configured threshold — check that the ESP32 sends at the expected interval, or adjust the threshold in Admin Panel → Categories → Edit.

Weather or PAR data is several hours late?

This is expected: LI-COR (HOBOnet) stations upload their measurements in batches, every 6 to 10 hours. The data appears at that point, at its original timestamp. A reading is only marked "Stale" past 12 hours.

A line is broken on a chart?

No data was received during that interval (sensor, power or network at the site). The platform does not join the points on either side of the gap.

I do not receive the e-mail alerts?

Alerts go to the administrators and to the users assigned to the site concerned whose "Receive offline alerts" box is ticked. Check your address in My Profile, your assignment to the site with an administrator, and your spam folder.